MIEB: Massive Image Embedding Benchmark

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Abstract

Image and image–text embedding models are widely used for retrieval, classification, clustering, and multimodal reasoning, yet evaluation remains fragmented and task-specific, obscuring how well a model generalizes across use cases. We introduce the Massive Image Embedding Benchmark (MIEB), a unified and reproducible evaluation suite spanning 130 tasks across 8 categories and 38 languages. MIEB extends the MTEB philosophy to vision by standardizing zero-shot evaluation across diverse modalities, languages, and task formats while keeping models frozen, with the sole exception of few-shot linear probes. The suite covers retrieval (including interleaved and instruction-aware settings), document understanding, linear probing, zero-shot classification, clustering, compositionality, vision-centric QA, and a new Visual STS that renders sentence pairs as images to diagnose visual text understanding. We further propose MIEB-lite, a 51-task subset derived from inter-task correlations that preserves diagnostic power while reducing runtime by 82%. Evaluations of representative encoders and MLLM-derived embeddings reveal complementary strengths: CLIP/SigLIP-style encoders excel at linear probing and zero-shot classification, while MLLM-based embeddings lead on OCR-heavy and multilingual tasks. Overall, no single model dominates all categories, underscoring the need for broad, standardized assessment.[1]

1. Introduction

Embeddings underpin modern multimodal systems, enabling efficient retrieval, zero-shot classification, and integration into large multimodal models (LMMs). Despite widespread use, evaluation is typically performed on narrow task subsets, which limits comparability and may misrepresent generalization. Inspired by the consolidation that the Massive Text Embedding Benchmark (MTEB) brought to text embeddings [2], we argue for a similarly comprehensive framework for image and image–text embeddings.

We present MIEB, a large-scale benchmark that unifies evaluation across modalities (image, text, interleaved), languages (38), and task types (eight categories). MIEB emphasizes under-evaluated abilities such as visual text understanding, interleaved inputs, compositionality, and multilinguality. Models are evaluated in a frozen setting to measure zero-shot generalization, with optional few-shot linear probes providing a controlled lens on linear separability.

Our design goals are coverage, realism, difficulty, contamination resistance, and diagnostic value. MIEB is built on the MTEB codebase to enable reproducibility and public tracking. A smaller MIEB-lite suite accelerates iteration while maintaining high agreement with full-benchmark rankings.

Empirically, we observe that: (1) No model leads across all categories; (2) CLIP-family encoders [3] and recent variants often achieve strong linear probing and zero-shot classification; (3) embeddings derived from MLLMs (e.g., E5-V [1]) tend to excel on OCR- and multilingual-heavy tasks; and (4) document understanding, interleaved retrieval, and compositionality remain challenging.

Contributions. MIEB offers: (i) a unified, frozen-evaluation suite spanning eight categories and 38 languages; (ii) a taxonomy that separates retrieval, linear separability, clustering structure, compositionality, and visual text understanding; (iii) a standardized protocol with fixed seeds and public manifests; (iv) Visual STS for probing OCR-centric capabilities in an embedding-first way; and (v) MIEB-lite, a reduced subset with strong rank agreement for fast iteration. Importantly, MIEB does not evaluate end-to-end generation, dense localization, or safety/harms; it measures embedding quality for retrieval- and classification-style objectives.

2. Related Work

Text embedding benchmarks such as MTEB [2] established the value of broad, standardized evaluation. For vision–language models, CLIP [3] catalyzed zero-shot image classification and retrieval, but many evaluations remain narrow. Compositionality datasets (e.g., Winoground [4] and ARO [6]) reveal persistent gaps in visio-linguistic reasoning. Community efforts track LMM performance across diverse tasks [5]. MIEB complements these by focusing on frozen embedding evaluation across a wide task taxonomy

[1] Leaderboard and code: https://hf.co/spaces/mteb/leaderboard

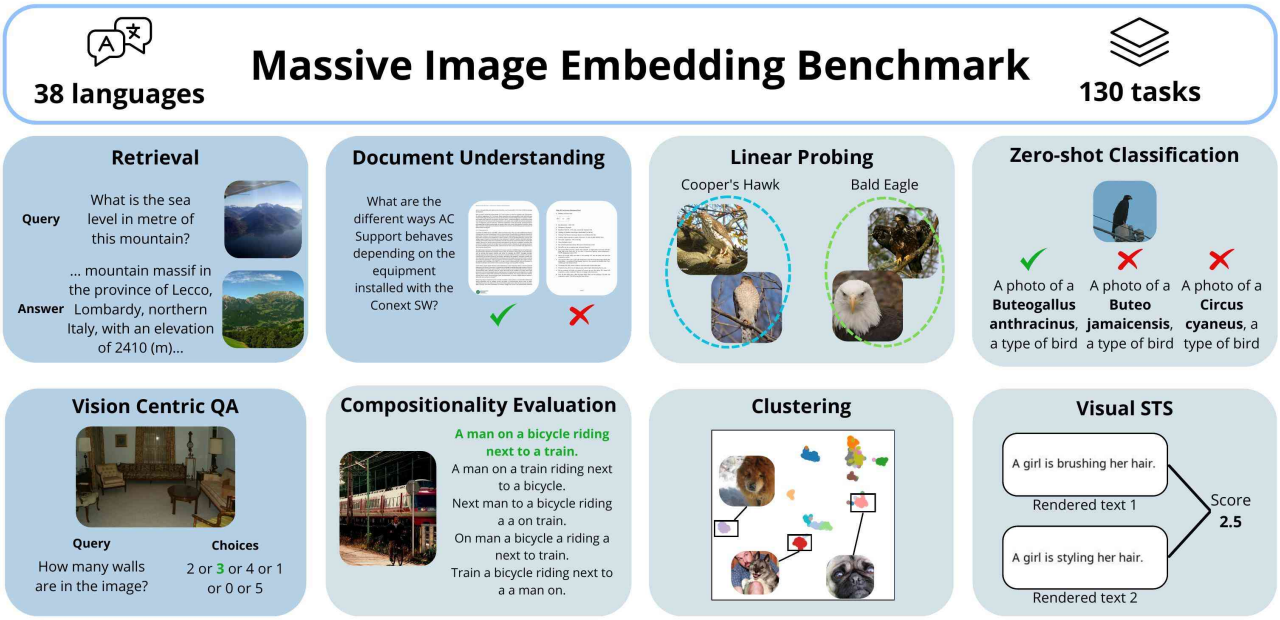

Figure 1. Overview of MIEB task categories with examples.

and by introducing Visual STS to probe visual text understanding.

3. Benchmark Design

MIEB spans 130 tasks grouped into eight categories: Retrieval (including interleaved, multilingual, and instruction-aware), Document Understanding, Linear Probing, Zero-shot Classification, Clustering, Compositionality, Vision-centric QA, and Visual STS. Tasks cover 38 languages and multiple modality pairings (image, text, interleaved). The overview table summarizes categories, counts, and modalities.

Task Definition and Inputs. Retrieval tasks include image–image, image–text, text–image, and interleaved combinations (e.g., it–t). Document understanding is evaluated via retrieval-style tasks using document pages with OCR content. Classification is probed via zero-shot prompts and few-shot linear probes. Clustering assesses the structure of embedding space, while compositionality tasks stress sensitivity to attributes, relations, and order. Vision-centric QA is cast as retrieval against candidate answers or evidence.

Visual STS. We render standard STS sentence pairs (STS12–17, STS-b) as images and compute Spearman correlation between cosine similarities and ground-truth similarities. This isolates visual text understanding in encoders, complementing text-only STS.

4. Data and Evaluation Protocol

Task sourcing. We curate public datasets emphasizing modality diversity (i, t, it), interleaving, multilingual coverage, and difficulty. Examples include multilingual retrieval (e.g., xFlickr&CO, XM3600, WIT), compositionality (ARO, Winoground), and standard classification sets (e.g., ImageNet-1k and common fine-grained benchmarks). Document understanding tasks follow visually rich document retrieval settings.

Unification and preprocessing. We standardize formats using the MTEB framework [2], version datasets and models, and ensure metric consistency. For Visual STS, we render text pairs to images; for interleaved inputs, we use model-native encoders when available and otherwise combine modalities by summing normalized image and text embeddings.

Evaluation settings. Unless otherwise specified, models are frozen. Zero-shot classification uses prompt templates (e.g., "a photo of [class]"). Linear probes train a logistic regression with 16 shots per class (max 100 iterations). Retrieval metrics use nDCG@10 (or nDCG@5 for document-centric tasks), with recall@1/MAP@5 for specific tasks. Clustering uses k-means with k equal to the number of ground-truth labels and is scored with NMI. Visual STS uses Spearman correlation.

5. Benchmark Validation

We validate MIEB along quality and diagnostic axes: (1) *Annotation quality and task validity* by reusing human an-

Task category	Example abilities assessed	# Tasks	# Languages	Modalities
Retrieval	cross-modal/-lingual matching	45	38	i-i; i-t; t-i; it-i; it-t; i-it; t-it; it-it; i-t
Document Understanding (Retrieval)	OCR abilities	10	2	t-i; i-t; it-t
Linear Probing (Classification)	information encoded	22	1	i-i; i-i
Clustering	embedding space consistency	5	1	i-i
Zero-shot Classification	cross-modal matching	23	1	i-t; i-t
Compositionality Evaluation (PairClassification)	reasoning with confounders	7	1	i-t; t-i
Vision-centric QA (Retrieval)	counting, object detection	6	1	it-t; it-i
Visual STS	OCR abilities	9	12	i-i
MIEB	all	130	38	all
MIEB-lite	all	51	38	all

Table 1. **An overview of MIEB tasks.** In brackets behind task categories, we denote the task type implementation in the code, e.g., our document understanding tasks use our retrieval implementation. We denote the modalities involved in both sides of the evaluation (e.g., queries and documents in retrieval; images and labels in zero-shot classification) with i=image, t=text.

notations and established splits; (2) *Distributional diversity* across languages, domains, and modality combinations; (3) *Difficulty calibration* via baseline sweeps; (4) *Decontamination checks* through versioning and dataset provenance; (5) *Metric robustness* via sensitivity analyses; and (6) *Agreement with external signals*. Notably, Visual STS shows a strong correlation with performance on generative LMM OCR/Chart evaluations (see Figure 2), indicating that vSTS is a useful proxy for OCR-centric capability.

To derive MIEB-lite, we compute task–task correlations using model performance, project tasks with UMAP, and cluster with HDBSCAN to identify a diverse subset that preserves ranking stability while excluding overly costly tasks. Runtime reductions are summarized in the runtime table.

6. Models and Experimental Setup

We evaluate representative families: (1) *Encoders* such as CLIP variants [3] and community-trained models; and (2) *MLLM-derived embeddings*, including E5-V [1] and related approaches. All evaluations use the unified MTEB-based pipeline, frozen models, and standardized metrics. For linear probing we report the average over multiple class-balanced draws.

7. Evaluation Protocol and Metrics

All models are evaluated in a frozen setting with a single pass to compute embeddings for each corpus and query set. Retrieval uses cosine similarity with L2-normalized vectors and reports nDCG@10 as the primary metric, with recall@1 and MAP@5 for designated benchmarks. Zero-shot classification follows the CLIP template protocol with class-name prompts and reports top-1 accuracy. Linear probing trains a logistic regression on 16 shots per class (up to 100 iterations) with class-balanced draws and reports average accuracy over five seeds. Clustering uses k-means with k set to the number of classes and reports NMI. For Visual STS we compute Spearman correlation on rendered sentence pairs. All runs fix seeds, evaluation code, and preprocessing; we release exact task manifests to support reproducibility.

8. Reproducibility and Release

We release: (i) machine-readable task manifests with dataset versions and license notes; (ii) a reference Docker/Conda environment with pinned packages; (iii) deterministic evaluation scripts with fixed seeds and canonical pooling/normalization; and (iv) per-model configuration files specifying resize/crop policies and maximum resolution. All reported runs use a single forward pass per corpus/query set and no test-time augmentation to ensure comparability. To aid downstream use, we also provide slice definitions for every category and a lightweight CLI that reproduces MIEB-lite with a single command. The public leaderboard reports metric definitions alongside scores and links to code, manifests, and model cards, supporting transparent and repeatable evaluation.

9. Benchmark Validation

We validate reliability via: (i) baseline sanity checks showing non-trivial headroom, (ii) data QA and near-duplicate removal, (iii) decontamination across splits, and (iv) metric and seed sensitivity analyses. Findings are stable under Recall@1 vs. nDCG@10, and Visual STS correlates with human judgments. Known weaknesses include residual overlap risks for widely circulated images and uneven coverage for low-resource scripts.

10. Benchmark Results

Overall results. The category-wise and overall scores for the top 20 models on MIEB (130 tasks) and MIEB-lite (51 tasks) are shown below. Overall, no single model dominates all categories. Voyage-multimodal-3 leads in the mean over

all tasks, while CLIP/SigLIP-style encoders achieve strong zero-shot classification and linear probing performance.

Multilingual retrieval. On multilingual retrieval across 38 languages, MLLM-based embeddings achieve the highest average scores and the lowest variance across language subsets, with E5-V showing particularly strong performance.

Visual STS and OCR-centric ability. E5-V approaches text-only STS performance on our Visual STS, indicating strong visual text understanding.

Efficiency: MIEB vs. MIEB-lite. MIEB-lite mirrors full-benchmark rankings while offering substantial runtime savings. For example, we observe an 82.4% reduction for an 8B-parameter MLLM embedding and 72.9% for a base CLIP encoder.

11. Benchmark Design and Protocol

Scope. MIEB evaluates frozen image and image–text embeddings across eight categories with standardized input/output formats; optional few-shot linear probes assess linear separability.

Taxonomy. Tasks cover retrieval (including interleaved and multilingual), document understanding, zero-shot classification, linear probing, clustering, compositionality, vision-centric QA, and Visual STS. An overview table summarizes the suite.

Metrics. We report accuracy for classification, NMI/AMI for clustering, and nDCG@10/Recall@k for retrieval; category means and a macro-average enable diagnostic comparisons.

12. Evaluation Protocol

We evaluate models frozen with a unified encoding API and fixed random seeds. For image–text inputs we use CLIP-style prompts with language variants; interleaved inputs concatenate modalities per dataset without instruction tuning. Embeddings use the model's native pooling (CLS/global token or mean). Linear probing uses L2-regularized logistic regression with 16 shots per class, early stopping on a held-out split, and macro accuracy for class-imbalanced tasks. Retrieval uses nDCG@10 and Recall@{1,5,10}; clustering uses NMI/AMI; we report category means and a macro average. We version models, tokenizers, and preprocessing, and apply simple decontamination filters when available. To assess stability, we repeat three runs for a subset of tasks and confirm consistent rankings within normal variance.

13. Analysis

Category sensitivity. Encoders optimized for image–text contrastive learning excel at label-aligned zero-shot classification and benefit strongly from linear probing, suggesting linearly separable features for common object categories. In contrast, MLLM-derived embeddings better capture multilingual and OCR-centric signals that drive document understanding and cross-lingual retrieval.

Compositionality and interleaving. Across compositionality stress tests and interleaved retrieval, most models remain far from robust performance, consistent with prior findings on Winoground and ARO [4, 6]. This indicates room for improvement in representing attributes, relations, and order.

Embedding space structure. Clustering analyses on fine-grained subsets (e.g., ImageNet Dog15) indicate meaningful but imperfect separation (Figure 5), aligning with NMI scores in our clustering category.

Ranking stability. Figure 6 shows strong agreement between full MIEB and MIEB-lite; Kendall $\tau > 0.9$ with only minor inversions among closely scoring CLIP variants, supporting MIEB-lite for fast iteration without masking material differences.

Shot efficiency. Consistent with Figure 4, linear probes yield monotonic gains with diminishing returns beyond 16 shots; smaller encoders benefit proportionally more than larger ones, but ordering across families remains largely unchanged. These patterns reinforce our choice of a modest 16-shot protocol for comparability and cost.

Cluster structure. Figure 7 qualitatively illustrates how different encoders organize categories in embedding space. We observe clear separation for coarse-grained domains but partially overlapping neighborhoods for fine-grained classes and multilingual text-in-image categories. These patterns are consistent with retrieval results and indicate that cluster compactness alone is not a sufficient success criterion: models can achieve high retrieval while retaining ambiguous boundaries in challenging subdomains. The benchmark's slice-wise reporting helps distinguish genuine semantic separation from artifacts of specific datasets or preprocessing.

14. Discussion

MIEB provides a comprehensive, reproducible view of image and image–text embeddings. It should be used to diagnose strengths and weaknesses across categories and to select encoders for downstream systems, including as front-ends to LMMs. MIEB is not a claim of comprehensive coverage: it primarily measures retrieval-style alignment, classification separability, cluster structure, visual text understanding, and compositional sensitivity across a wide but finite set of domains and languages. Known limitations include incomplete coverage of non-photographic domains, remaining annotation subjectivity in source datasets, potential contamination risks inherent to public benchmarks, and the need for broader stress tests of robustness and safety-critical settings. We encourage using MIEB-lite for rapid

MIEB Full (130 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (45)**	**Clus. (5)**	**ZS. (23)**	**LP. (22)**	**Cmp. (7)**	**VC. (6)**	**Doc. (10)**	**vSTS (en) (7)**	**Rtrv. (m) (3 (55))**	**vSTS (x&m) (2 (19))**	**Mean (en) (125)**	**Mean (m) (130)**
Voyage-multimodal-3	MLLM	38.8	82.4	58.2	71.3	43.5	48.6	**71.1**	**81.8**	58.9	**70.4**	**62.0**	**62.5**
E5-V	MLLM	34.0	70.0	50.0	74.5	**46.3**	51.9	62.7	79.3	**66.6**	46.3	58.6	58.2
siglip-so400m-patch14-384	Enc.	40.8	82.1	**70.8**	84.6	40.4	46.3	56.4	68.0	40.2	41.4	61.2	57.1
siglip-large-patch16-384	Enc.	39.9	79.9	68.0	83.7	39.7	45.4	53.3	69.5	51.1	39.8	59.9	57.0
siglip-large-patch16-256	Enc.	38.8	82.1	67.7	82.5	40.8	44.9	39.4	67.4	49.8	38.1	57.9	55.2
siglip-base-patch16-512	Enc.	38.1	74.7	64.1	80.9	37.5	53.2	52.1	67.7	43.2	38.1	58.5	54.9
CLIP-ViT-bigG-14-laion2B	Enc.	**41.5**	85.6	69.4	83.6	42.4	43.2	43.2	70.9	28.0	34.5	60.0	54.2
siglip-base-patch16-384	Enc.	37.7	76.3	64.1	80.6	38.5	52.8	45.0	67.0	42.5	37.5	57.8	54.2
EVA02-CLIP-bigE-14-plus	Enc.	40.1	**92.4**	70.8	**86.0**	45.7	39.4	32.3	72.0	27.8	28.2	59.8	53.5
CLIP-ViT-L-14-DataComp.XL	Enc.	38.1	86.4	68.4	82.0	39.1	52.3	38.6	69.9	23.8	35.8	59.4	53.4
siglip-base-patch16-256(m)	Enc.	35.6	74.6	61.2	78.9	38.1	51.3	26.4	65.5	59.2	40.3	53.9	53.1
CLIP-ViT-H-14-laion2B	Enc.	39.7	83.9	67.5	82.5	42.0	45.8	40.4	65.5	25.5	33.9	58.4	52.7
CLIP-ViT-g-14-laion2B	Enc.	39.8	82.7	67.9	82.8	41.9	44.2	37.6	69.1	25.9	31.7	58.3	52.4
EVA02-CLIP-bigE-14	Enc.	39.0	89.4	69.3	84.5	42.4	43.6	31.6	68.8	25.5	28.3	58.6	52.2
siglip-base-patch16-256	Enc.	36.6	75.2	63.1	79.7	39.5	52.2	31.7	66.2	41.3	34.4	55.5	52.0
siglip-base-patch16-224	Enc.	36.3	74.5	62.6	79.3	39.8	51.1	26.2	64.3	41.2	33.5	54.3	50.9
CLIP-ViT-L-14-laion2B	Enc.	38.0	83.5	65.8	81.2	40.8	45.9	36.3	65.8	23.0	26.0	57.2	50.6
VLM2Vec-LoRA	MLLM	27.7	72.6	46.3	62.0	34.6	62.0	49.7	72.6	34.9	42.2	53.4	50.5
VLM2Vec-Full	MLLM	27.6	70.7	46.3	62.0	35.4	**62.1**	49.8	72.6	35.0	42.2	53.3	50.4
clip-vit-large-patch14	Enc.	33.7	76.4	62.1	80.1	44.8	44.1	38.0	64.5	20.2	35.1	55.4	49.9
MIEB-lite (51 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (11)**	**Clus. (2)**	**ZS. (7)**	**LP. (8)**	**Cmp. (6)**	**VC. (5)**	**Doc. (6)**	**vSTS (en) (2)**	**Rtrv. (m) (2 (47))**	**vSTS (x&m) (2 (19))**	**Mean (en) (47)**	**Mean (m) (51)**
Voyage-multimodal-3	MLLM	33.2	76.6	48.6	69.3	35.8	50.0	**63.5**	**84.2**	49.0	**70.4**	**57.7**	**58.1**
siglip-so400m-patch14-384	Enc.	32.4	75.9	73.8	78.8	32.8	48.0	46.9	69.6	35.4	41.4	57.3	53.5
siglip-large-patch16-384	Enc.	31.9	75.2	71.3	77.7	32.1	46.8	44.9	69.6	43.5	39.8	56.2	53.3
E5-V	MLLM	26.9	51.7	36.2	70.6	**39.4**	52.6	56.0	81.2	**58.3**	46.3	51.8	51.9
siglip-large-patch16-256	Enc.	31.0	76.5	70.3	76.3	33.4	46.5	31.9	67.6	42.6	38.1	54.2	51.4
CLIP-ViT-bigG-14-laion2B	Enc.	34.2	80.8	72.4	77.8	35.0	43.0	35.5	73.4	26.2	34.5	56.5	51.3
siglip-base-patch16-512	Enc.	30.8	69.7	66.3	74.6	29.7	55.5	42.6	67.1	34.8	38.1	54.5	50.9
EVA02-CLIP-bigE-14-plus	Enc.	**35.2**	**87.3**	**74.0**	**80.0**	38.9	38.8	26.2	73.7	26.0	28.2	56.8	50.8
siglip-base-patch16-384	Enc.	30.6	72.2	66.0	74.4	31.0	55.1	37.1	66.9	34.5	37.5	54.1	50.5
CLIP-ViT-L-14-DataComp.XL	Enc.	31.0	80.4	69.4	75.3	31.6	54.9	30.8	72.5	22.6	35.8	55.7	50.4
CLIP-ViT-H-14-laion2B	Enc.	32.8	79.3	69.4	76.8	34.8	46.8	33.7	68.3	23.9	33.9	55.2	50.0
EVA02-CLIP-bigE-14	Enc.	34.3	86.7	73.0	78.3	35.1	44.4	25.1	69.9	23.9	28.3	55.9	49.9
siglip-base-patch16-256(m)	Enc.	28.2	68.2	63.2	73.4	30.7	53.3	22.9	63.7	52.9	40.3	50.4	49.7
CLIP-ViT-g-14-laion2B	Enc.	33.5	76.8	69.6	77.3	34.7	45.0	29.9	71.6	24.2	31.7	54.8	49.4
siglip-base-patch16-256	Enc.	29.5	69.6	65.6	73.6	32.2	54.4	25.0	66.1	33.5	34.4	52.0	48.4
CLIP-ViT-L-14-laion2B	Enc.	31.1	76.4	67.8	75.9	33.6	46.9	28.7	68.7	21.4	26.0	53.6	47.6
clip-vit-large-patch14	Enc.	26.7	71.3	63.8	74.5	**39.4**	44.9	29.4	69.4	19.8	35.1	52.4	47.4
siglip-base-patch16-224	Enc.	29.3	68.4	65.0	73.5	32.5	53.0	20.9	64.2	33.6	33.5	50.8	47.4
CLIP-ViT-B-16-DataComp.XL	Enc.	28.3	73.6	61.9	73.2	31.4	56.9	22.7	69.7	19.9	28.5	52.2	46.6
VLM2Vec-LoRA	MLLM	21.0	66.3	32.1	64.8	29.4	**65.3**	42.7	70.9	24.8	42.2	49.1	46.0

Table 2. **MIEB results broken down by task categories for the top 20 models.** We provide averages of both English and multilingual tasks. Models are ranked by the Mean (m) column. Shortcuts are x=Crosslingual, m=Multilingual, en=English, and task categories from Figure 1. We refer to the leaderboard for the latest version: https://hf.co/spaces/mteb/leaderboard

Model Name	**xFlickr&CO**		**XM3600**		**WIT**		**avg.**	
	avg.	var.	avg.	var.	avg.	var.	avg.	var.
E5-V	**90.8**	**0.1**	**74.8**	3.5	**57.3**	0.6	**74.3**	1.4
SigLIP	80.4	1.2	65.6	5.3	54.4	1.3	66.8	2.6
VISTA (m3)	65.3	0.2	48.5	**2.0**	49.3	**0.4**	54.4	**0.9**
VLM2Vec	63.8	3.8	27.0	4.7	31.7	2.5	40.8	3.6
Open-CLIP	35.9	9.3	20.5	6.0	37.8	6.5	31.4	7.3
EVA02-CLIP	35.6	9.4	20.1	6.0	37.4	6.4	31.0	7.2

Table 3. **Performance of models on multilingual retrieval tasks across 38 languages.** We compute the average performance across languages (avg) and the respective variance (var). We take the best variant from each top-6 model family.

	12	**13**	**14**	**15**	**16**	**17**	**b**	**avg.**
STS*	80.0	89.9	85.7	89.1	85.9	87.9	83.5	86.0
v-STS (ours)	73.2	78.2	74.9	84.2	79.5	85.8	79.4	79.3

Table 4. **E5-V performance on regular STS and our Visual STS.** *: numbers from Jiang et al. [1]. Columns are STS12-17 and STS-b.

iteration and the full MIEB for conclusive reports.

Threats to validity. While we control seeds, metrics, and preprocessing, residual confounders remain. Some

Model Name	# Params (M)	Runtime (NVIDIA H100 GPU hours)		
		MIEB	MIEB-lite	Reduction %
E5-V	8360	264.0	46.4	82.4% ↓
CLIP (base-patch32)	151	16.6	4.5	72.9% ↓

Table 5. **MIEB vs. MIEB-lite runtime comparison.**

tasks inherit long-tail label noise, and multilingual coverage is uneven across scripts. Publicly released models may have unseen exposure to subsets of our corpora despite version pinning; our decontamination checks reduce but cannot eliminate this risk. Moreover, slice granularity trades off interpretability and statistical power; we therefore report both per-slice and category means and encourage caution when over-interpreting small deltas. Finally, MIEB evaluates embedding utility rather than end-to-end generation or spatial grounding; high scores do not imply instruction following, dense localization, or safety. We also record hardware, CUDA/cuDNN, BLAS, and tokenizer versions; small numerical drift may persist but leaves qualitative patterns and rankings stable.

15. Conclusion

We introduced MIEB, a large-scale benchmark for image and image–text embeddings spanning 130 tasks across eight categories and 38 languages. MIEB standardizes frozen evaluation and adds Visual STS for probing visual text understanding. Results across representative encoders and MLLM-derived embeddings show no universal winner but clear, complementary strengths. MIEB-lite accelerates evaluation while preserving diagnostic value, enabling more reliable and informative assessment of embedding models.

References

[1] Ting Jiang, Minghui Song, Zihan Zhang, Haizhen Huang, Weiwei Deng, Feng Sun, Qi Zhang, Deqing Wang, and Fuzhen Zhuang. E5-v: Universal embeddings with multimodal large language models. *arXiv preprint arXiv:2407.12580*, 2024. 1, 3, 5

[2] Niklas Muennighoff, Nouamane Tazi, Loic Magne, and Nils Reimers. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037, 2023. 1, 2

[3] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021. 1, 3

[4] Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross. Winoground: Probing vision and language models for visio-linguistic compositionality. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5238–5248, 2022. 1, 4

[5] Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, et al. Cambrian-1: A fully open, vision-centric exploration of multimodal llms. *arXiv preprint arXiv:2406.16860*, 2024. 1, 8

[6] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? In *International Conference on Learning Representations*, 2023. 1, 4

Task category	Example abilities assessed	# Tasks	# Languages	Modalities
Retrieval	cross-modal/-lingual matching	45	38	i-i; i-t; t-i; it-i; it-t; i-it; t-it; it-it; i-t
Document Understanding (Retrieval)	OCR abilities	10	2	t-i; i-t; it-t
Linear Probing (Classification)	information encoded	22	1	i-i; i-i
Clustering	embedding space consistency	5	1	i-i
Zero-shot Classification	cross-modal matching	23	1	i-t; i-t
Compositionality Evaluation (PairClassification)	reasoning with confounders	7	1	i-t; t-i
Vision-centric QA (Retrieval)	counting, object detection	6	1	it-t; it-i
Visual STS	OCR abilities	9	12	i-i
MIEB	all	130	38	all
MIEB-lite	all	51	38	all

Table 6. **An overview of MIEB tasks.** In brackets behind task categories, we denote the task type implementation in the code, e.g., our document understanding tasks use our retrieval implementation. We denote the modalities involved in both sides of the evaluation (e.g., queries and documents in retrieval; images and labels in zero-shot classification) with i=image, t=text.

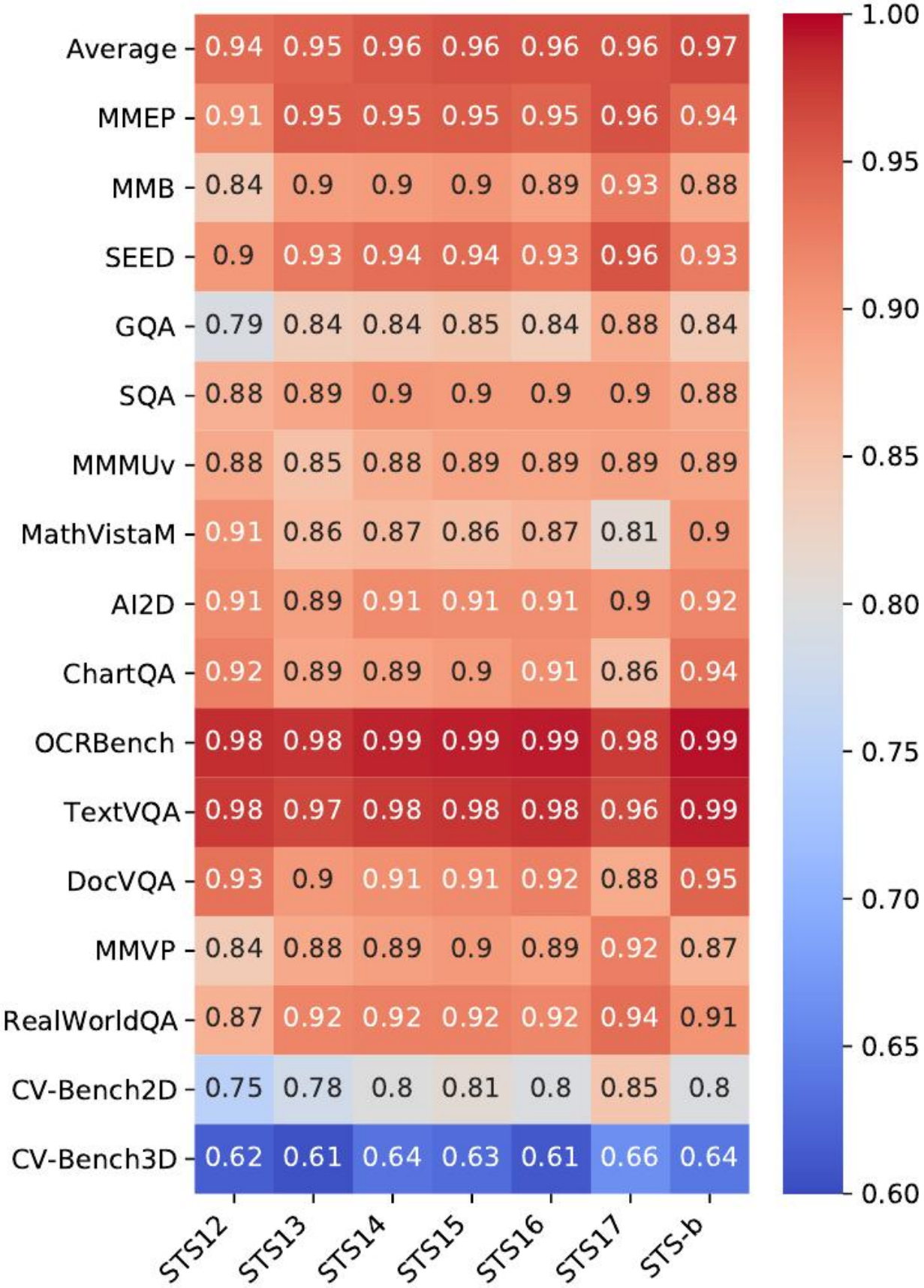

Figure 2. Correlation between performance on generative LMM benchmarks from [5] (y-axis) and Visual STS (x-axis).

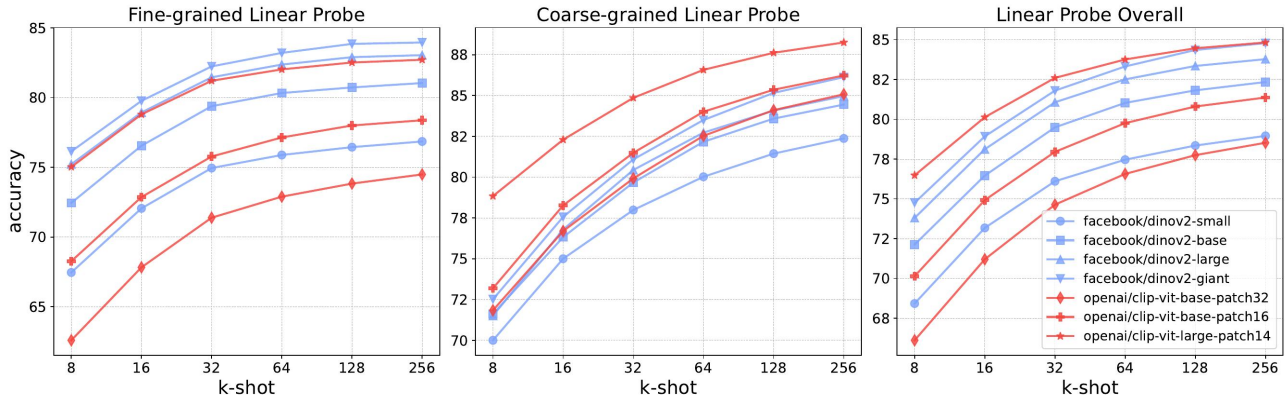

Figure 3. Linear probing performance across different shots k.

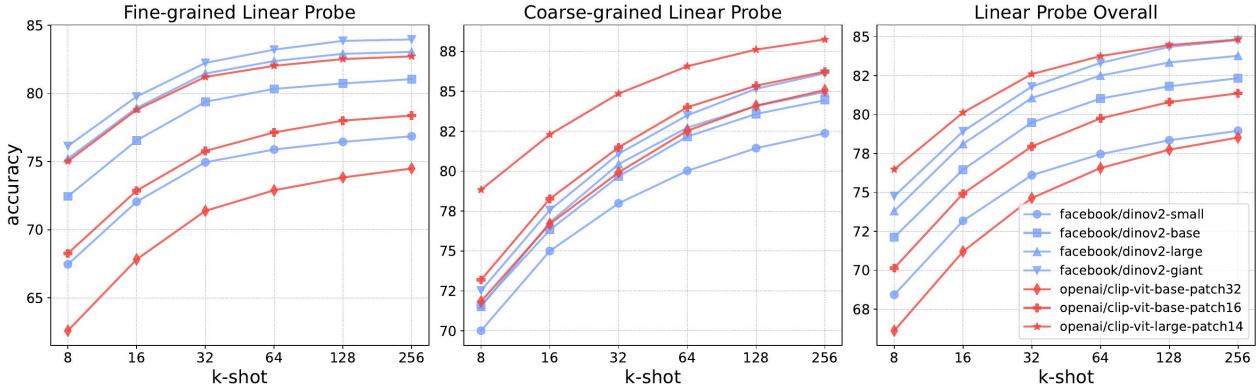

Figure 4. Shot-wise linear probing gains across representative tasks and models. Gains are monotonic with diminishing returns; we adopt 16-shot probes for comparability.

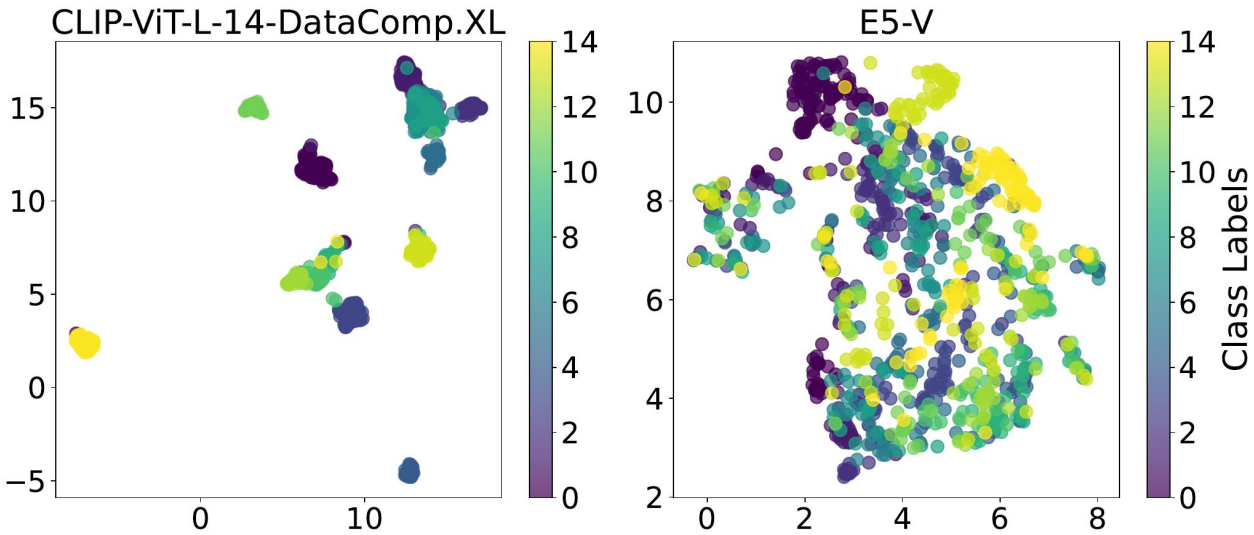

Figure 5. UMAP visualization of ImageNet Dog15 embeddings.

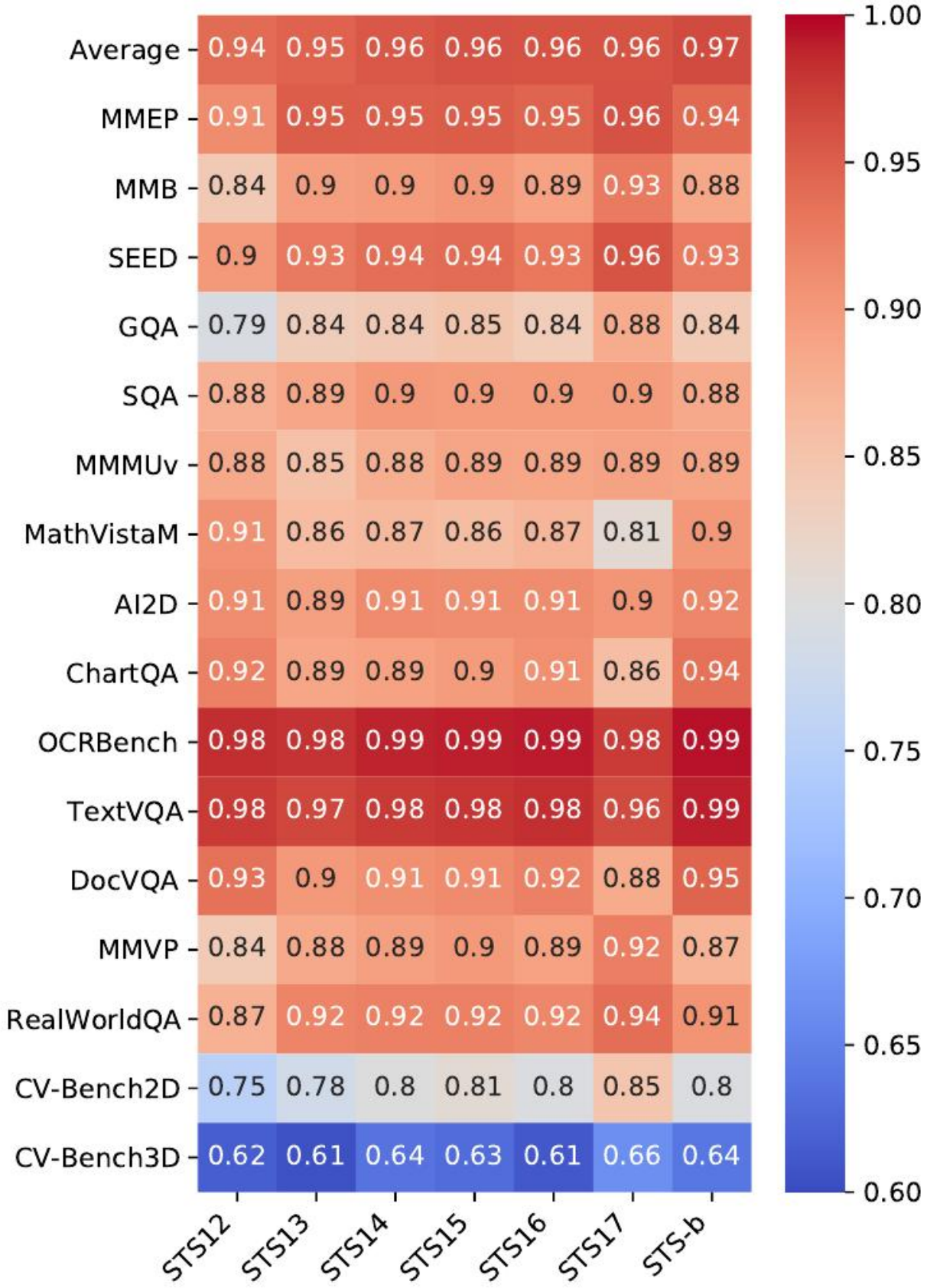

Figure 6. Rank correlation between full MIEB and MIEB-lite.

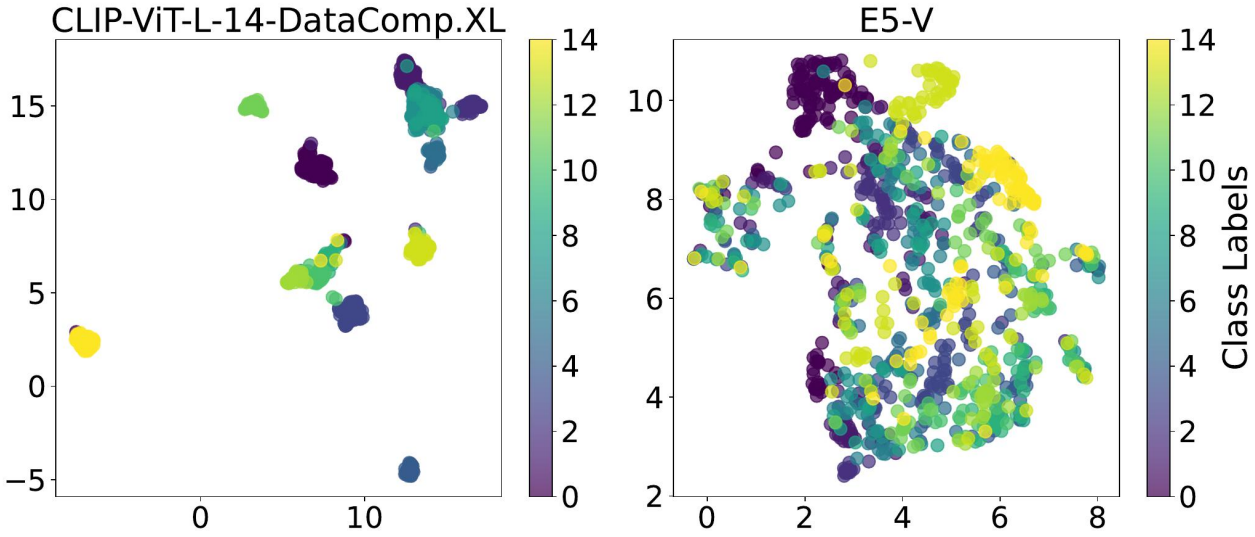

Figure 7. Qualitative cluster geometry across representative encoders. Coarse-grained categories form well-separated regions, while fine-grained and multilingual text-in-image classes exhibit overlapping boundaries.